

FITBUDDY

Project Documentation

Project Type: AI-Powered Fitness & Wellness Assistant

Documentation Phase

Project Name	FitBuddy
Domain	Health, Fitness & Wellness
Application Type	Web / Mobile-friendly Application
Main Purpose	Personalized workout, nutrition and fitness guidance
Target Users	Students, beginners and general fitness users

1. Introduction

FitBuddy is a fitness and wellness application designed to help users maintain a healthier lifestyle through personalized fitness guidance. The system can provide workout suggestions, basic nutrition guidance, progress tracking and motivational support based on user-provided information.

2. Problem Statement

Many users find it difficult to create a suitable workout routine, maintain consistency and monitor their progress. Generic fitness plans may not match an individual's goals, fitness level or available time. FitBuddy aims to provide a simple and personalized platform that brings these activities together in one place.

3. Objectives

- Create personalized workout plans based on user goals and fitness level.
- Provide simple exercise instructions and routine guidance.
- Allow users to record and monitor fitness progress.
- Provide basic nutrition and wellness suggestions.
- Improve motivation through reminders, achievements and progress information.

4. Scope of the Project

The project focuses on creating an easy-to-use fitness assistant. The initial version can include user registration/profile management, fitness goal selection, workout recommendations, exercise details, progress tracking and a dashboard. Advanced features such as wearable-device integration, professional trainer consultation and medical-grade recommendations may be considered as future enhancements.

5. Proposed System

The proposed FitBuddy system collects basic user preferences such as fitness goal, activity level and available workout time. The application processes these inputs and presents suitable fitness content. User activities and progress can be stored so that the dashboard can show useful summaries over time.

6. Major Modules

Module	Description
User Management	Registration, login and profile information.
Fitness Profile	Stores goals, fitness level, preferences and basic details.
Workout Planner	Generates or displays suitable workout routines.
Exercise Guide	Provides exercise names, instructions and routine details.
Nutrition & Wellness	Provides general healthy-lifestyle and nutrition guidance.
Progress Tracker	Records workouts and displays progress summaries.
Dashboard	Shows goals, activities, progress and useful recommendations.

7. Functional Requirements

- The system shall allow users to create and manage a fitness profile.
- The system shall allow users to select fitness goals.
- The system shall display appropriate workout recommendations.
- The system shall provide exercise information.
- The system shall record completed workouts and progress.
- The system shall display the user's progress through a dashboard.

8. Non-Functional Requirements

- Usability: The interface should be simple and easy to understand.
- Performance: Pages and recommendations should load efficiently.
- Security: User data should be protected using appropriate authentication and storage practices.
- Reliability: The system should handle normal user operations consistently.
- Responsiveness: The application should work well on different screen sizes.

9. Technology Stack

Layer	Suggested Technology
Frontend	HTML, CSS, JavaScript / React
Backend	Python Flask / Node.js
Database	MySQL / MongoDB / Firebase
AI Component	AI/ML model or API for personalized recommendations
Development Tools	VS Code, Git/GitHub

10. System Workflow

User Registration → Fitness Profile → Goal Selection → Recommendation Engine → Workout / Wellness Plan → Activity Tracking → Progress Dashboard

11. Data Flow Overview

The user enters profile and fitness information. The application validates the input and sends relevant information to the recommendation component. The resulting workout or wellness suggestions are displayed to the user. Completed activities are stored and later used to generate progress summaries.

12. Database Overview

A basic database design may contain the following entities:

- **User:** user ID, name, email and account information.
- **Fitness Profile:** goal, activity level, preferences and other basic inputs.
- **Workout:** workout ID, exercise list, duration and difficulty.
- **Progress:** date, completed workout, duration and progress values.
- **Recommendation:** user input, recommended activity and recommendation date.

13. Testing Plan

The application should be tested module-by-module and as a complete system. Important tests include registration/login validation, profile input validation, workout recommendation accuracy, progress-recording functionality, responsive UI testing and database operations.

14. Expected Outcome

The completed FitBuddy application is expected to provide users with a convenient platform for organizing workouts, receiving personalized fitness guidance and monitoring progress. The system should have a clear interface and provide useful recommendations based on the information supplied by the user.

15. Future Enhancements

- Integration with smartwatches and fitness trackers.
- Voice-based fitness assistance.
- More advanced AI personalization.
- Workout reminders and notifications.
- Progress charts and detailed analytics.
- Optional trainer or expert consultation features.

16. Conclusion

FitBuddy is designed as a practical fitness and wellness assistant that combines personalized recommendations, workout guidance and progress tracking in a single application. The project

demonstrates how web technologies and AI-based recommendations can be used to support users in maintaining consistent fitness habits.

Note: FitBuddy provides general fitness and wellness information and should not be treated as a substitute for professional medical advice.